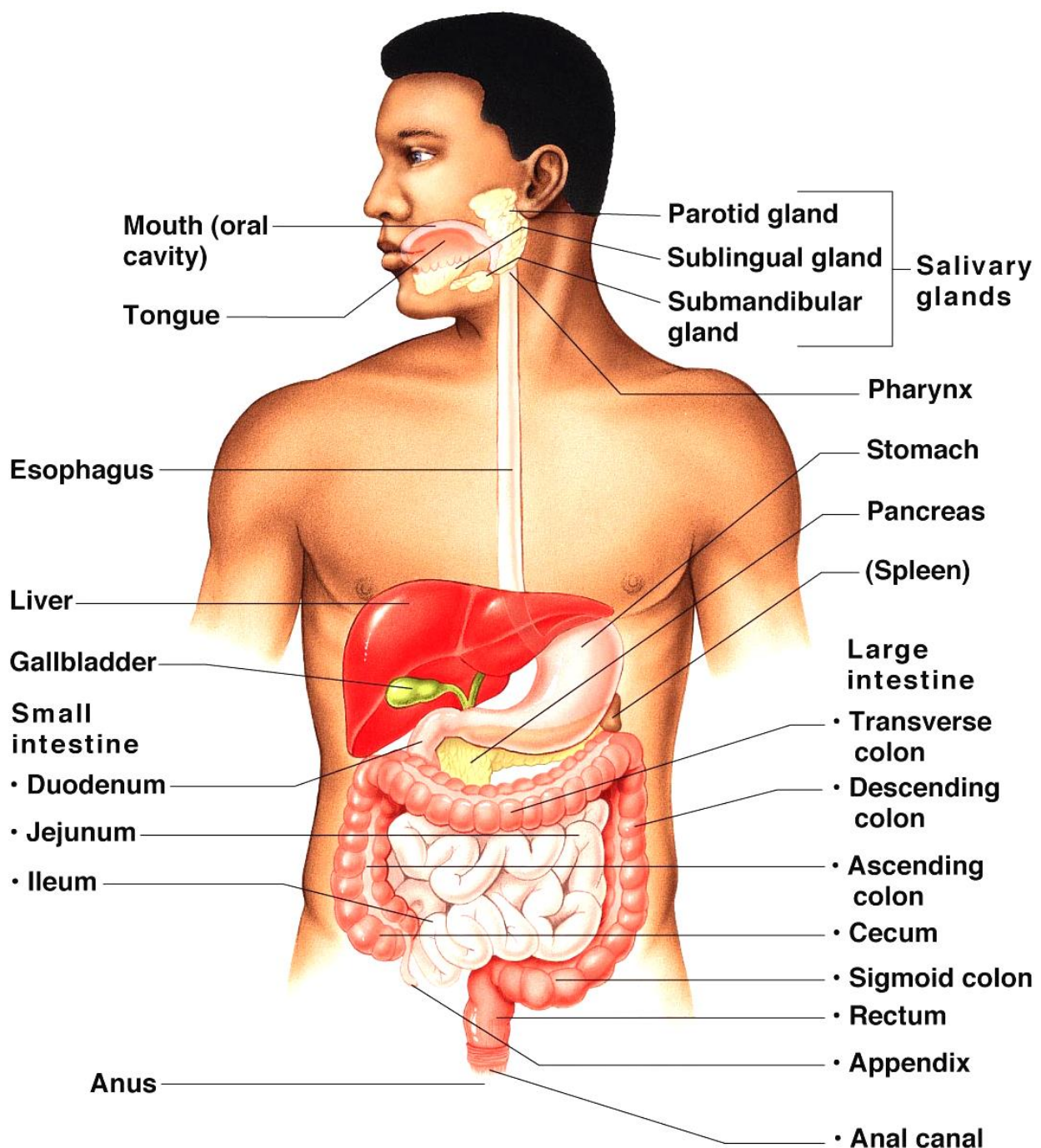


DIGESTIVE SYSTEM

* It **consists of** the following parts:

- 1- Mouth cavity and salivary glands.
- 2- Pharynx.
- 3- Oesophagus
- 4- Stomach.
- 5- Small intestine
- 6- Large intestine.
- 7- Liver and biliary system.
- 8- Pancreas.



Mouth (oral) Cavity

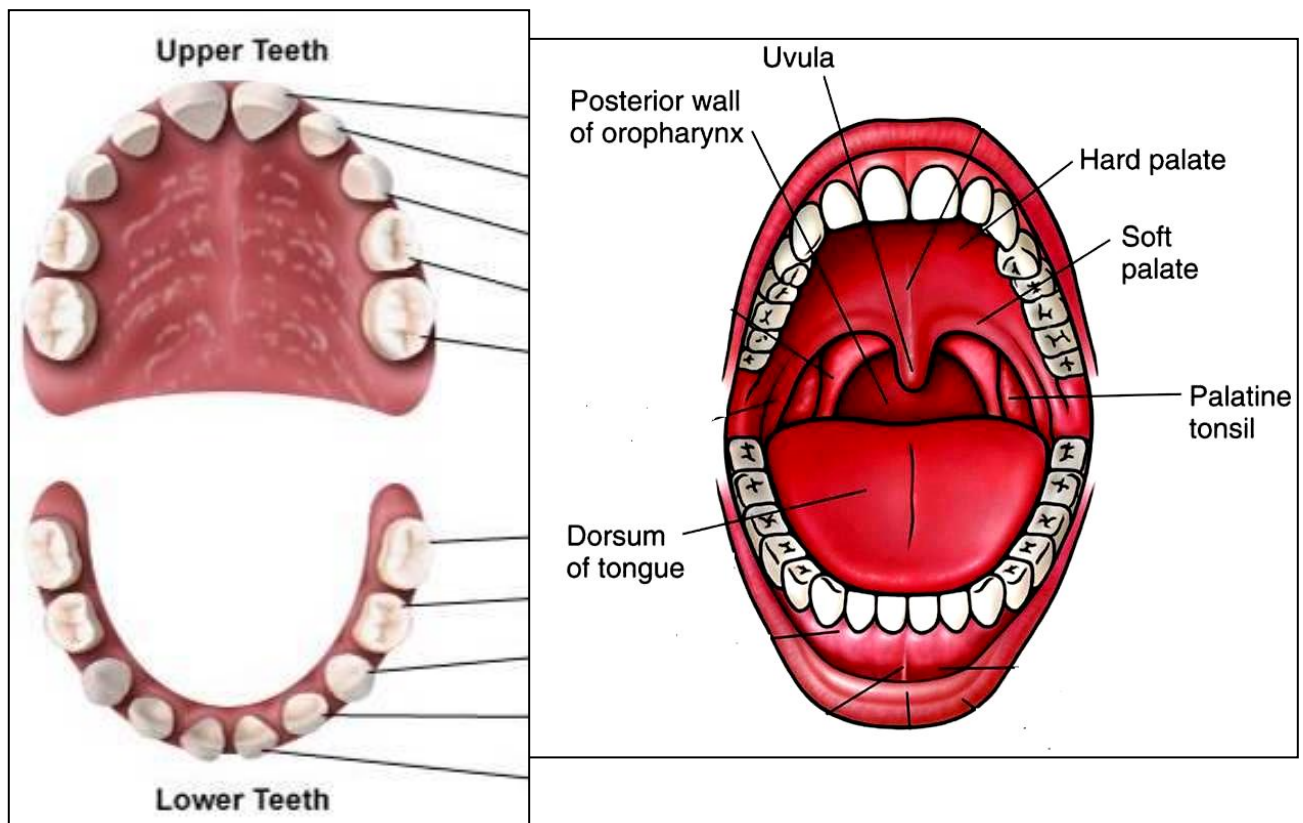
* **Parts (division):** It has 2 parts

A- Vestibule of the mouth cavity: It is U-shaped space which lies outside the teeth and gums. It is bounded by the teeth and gums internally and the cheek and lips externally.

B- Mouth cavity proper: Boundaries:

1. **Anteriorly and laterally:** teeth and gums (الثة).
2. **Posteriorly:** It is continuous with the oropharynx.
3. **Superiorly (roof):** the palate (سقف تجويف الفم).
4. **Inferiorly (floor):** Muscles covered by mucous membrane and the tongue.

* **Contents of the mouth cavity:** Teeth-tongue-openings of the salivary glands.



**Milk (Deciduous) Teeth
& Permanent teeth**

Mouth cavity

The Tongue

- It is a muscular organ covered by mucous membrane.
- It has **2 surfaces**:

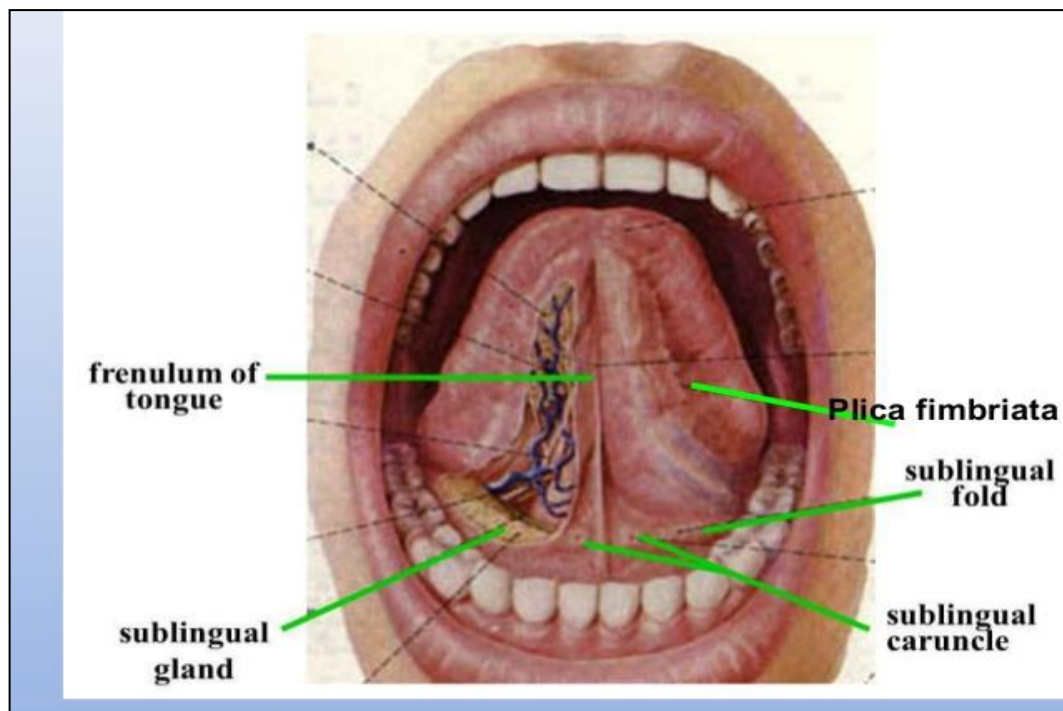
A- Dorsal surface: It is rough surface and divided by V-shaped sulcus named ***sulcus terminalis*** into anterior 2/3 and posterior 1/3.

Anterior 2/3 (oral part)	Posterior 1/3 (pharyngeal part)
▪ Directed upwards ▪ Lies in the oral (mouth) cavity	▪ Directed backwards ▪ Lies in the oropharynx

B- Ventral surface: It is smooth and is connected to the floor of the mouth by mucosal fold called ***frenulum linguae***.

* **Functions** of the tongue:

- 1- Chewing.
- 2- Speech.
- 3- Taste.
- 4- Swallowing.



Teeth

* Two types:

- I. **Milk (deciduous) teeth:** Temporary teeth present in the children, 20 in number 10 above and 10 below.
- II. **Permanent teeth:** They are 32 teeth, 16 above and 16 below, arranged in each half of jaw as follows: 2 incisors قواطع, canine ناب, 2 premolars, 3 molars.

Salivary Glands

1- Parotid gland

- It is the **largest** salivary gland, wedge-shaped, lies on the side of the face, anterior and inferior to the external auditory meatus.
- It is **formed** of main part, accessory part and has a duct.
- Its **duct** opens in the vestibule of the mouth cavity.

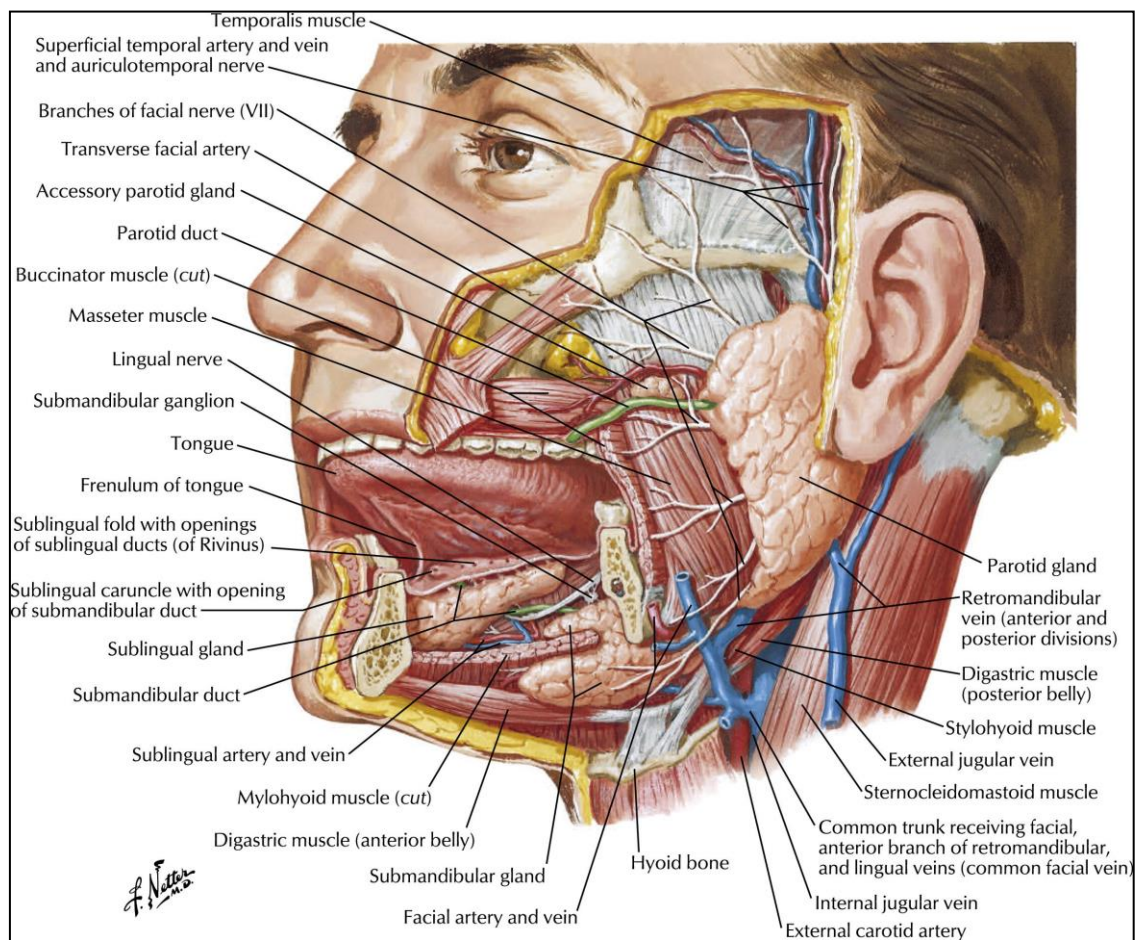
2- Submandibular gland

- It **lies** deep to and below the mandible.
- It is formed of superficial part, deep part and duct.
- Its **duct** opens in the floor of the mouth in the sublingual papilla on either side of the frenulum of tongue.

3- Sublingual gland

- It is the **smallest** salivary gland.
- It **lies** under the mucous membrane of the floor of the mouth.
- It has multiple **ducts** that open in the floor of the mouth.

Salivary glands



Pharynx

- * Funnel-shaped muscular tube, **lies** behind the nasal cavity, oral cavity and larynx.
- * It **begins** at the base of skull and **ends** at the level of lower border of C6 vertebrae by forming the esophagus.
- * It **lies** in front of the upper 6 cervical vertebrae and is **divided into 3 parts**:

1-Nasopharynx

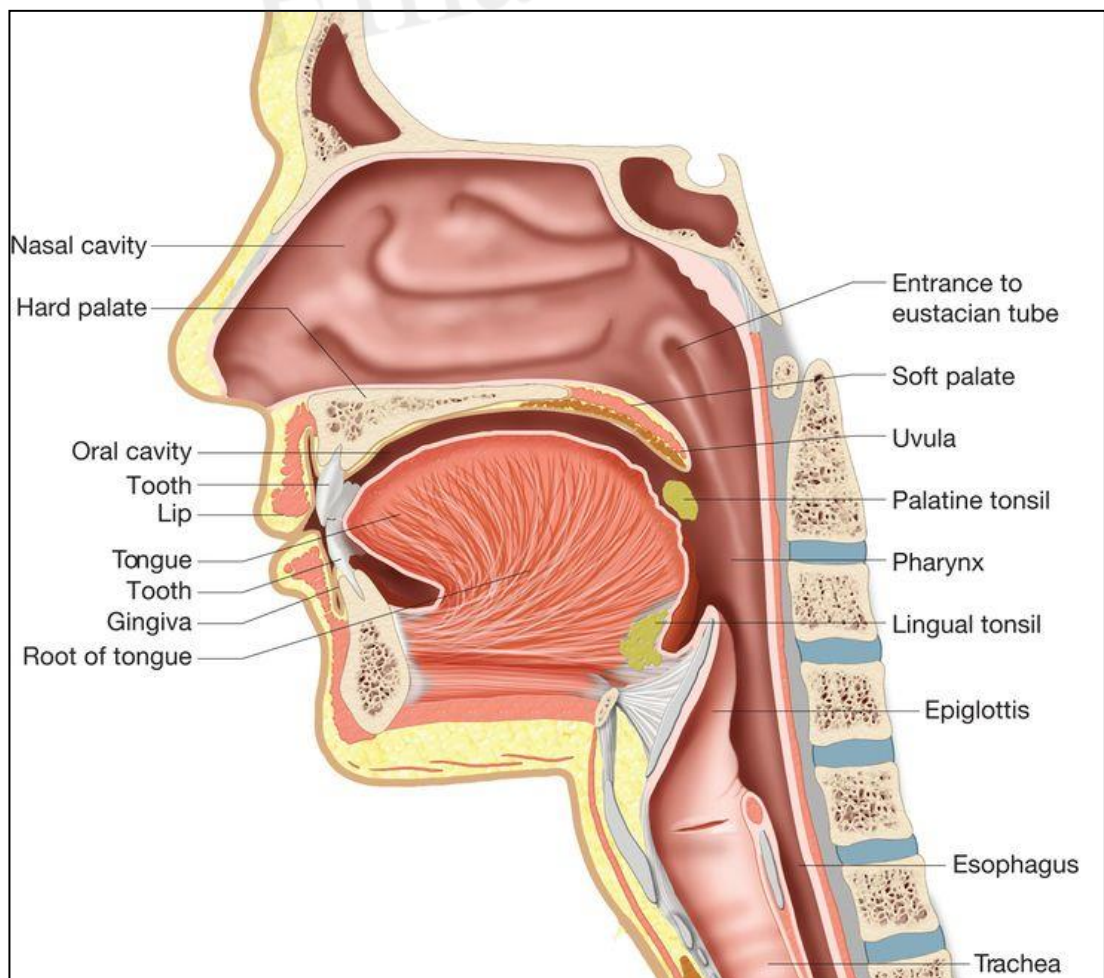
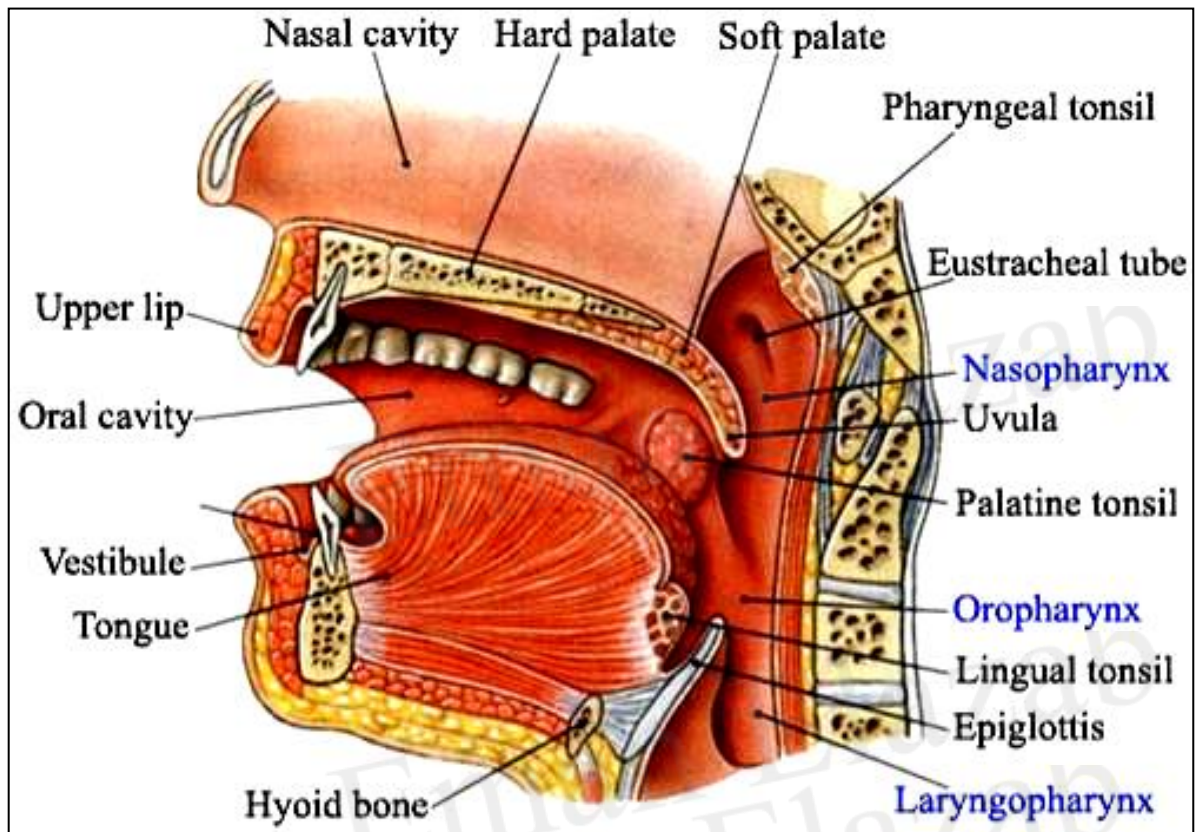
- It **lies** behind the nasal cavity, above the soft palate and acts as a passage for air only.
- Its upper part contains the **nasopharyngeal (pharyngeal) tonsil** while the lower part shows the opening of the pharyngotympanic (Eustachian) tube.
- It is **separated** from the oropharynx by the soft palate.
- It is a ***passage for*** air.

2-Oropharynx

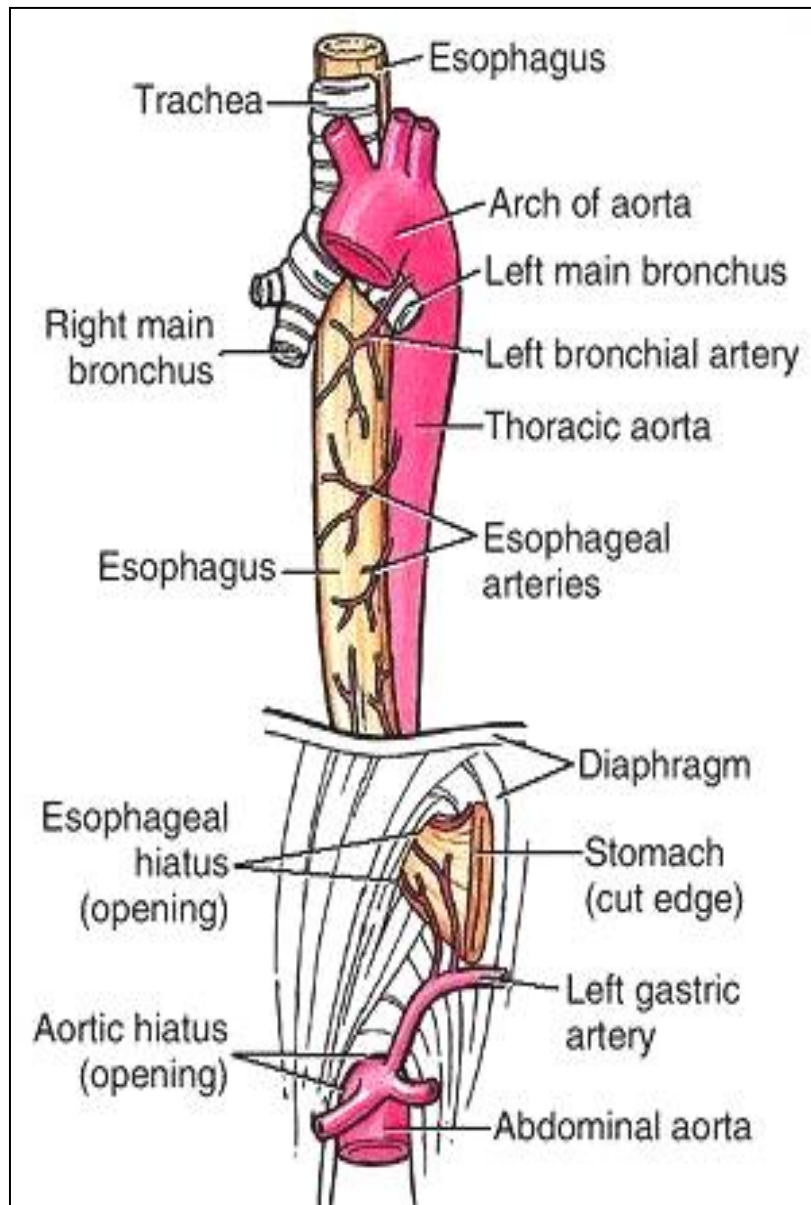
- It **lies** behind the oral (mouth) cavity.
- It is **separated** from the nasopharynx by the soft palate and separated from laryngopharynx by the upper border of epiglottis.
- It acts as a ***passage for*** food and air.
- Its lateral wall contains the **palatine tonsil**.

3-Laryngopharynx

- It **lies** behind the larynx and acts as a ***passage for*** food only.



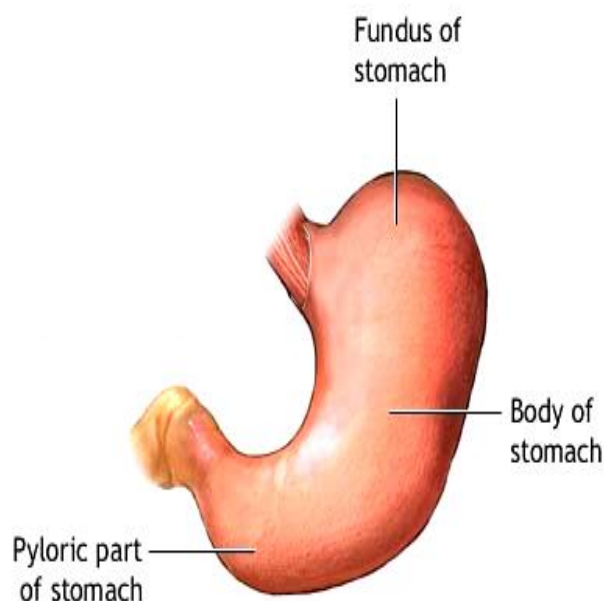
Esophagus

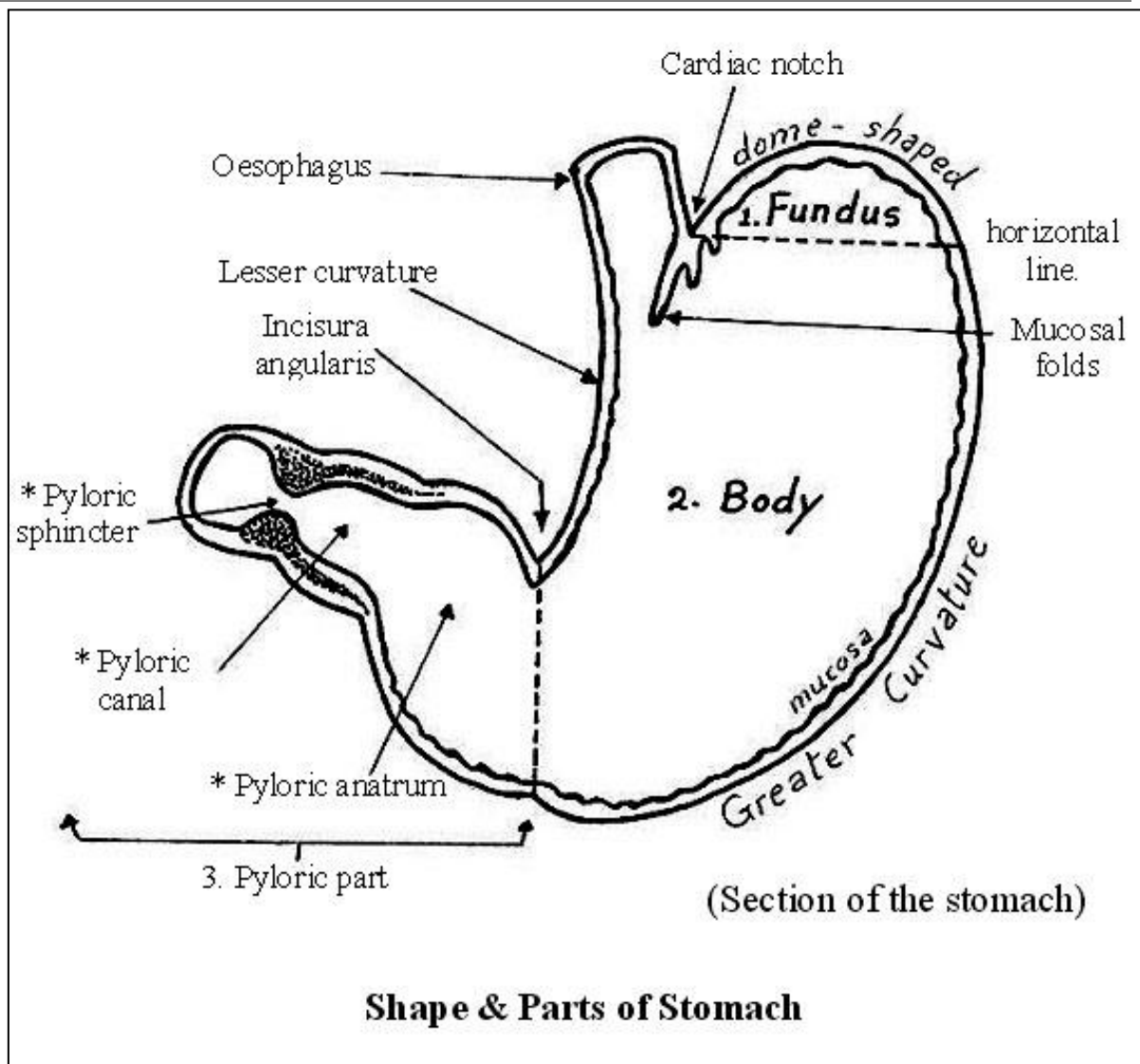


- * It is a muscular tube 25 cm long,
- * It **begins** in the neck as continuation of the pharynx.
- * It descends for a short course in the neck, then it enters the thoracic cavity, then it passes through the diaphragm to enter the abdomen where it ends in the cardiac end of the stomach.
- * The esophagus has **3 parts**: cervical part, thoracic part and abdominal part.

Stomach

- * It is the most dilated part of the digestive tract.
- * **Site:** It lies in the upper part of the abdominal cavity behind the left lobe of liver and anterior wall of abdomen.
- * **Shape:** is variable according the position of the body and according to the condition of the stomach if it is empty or filled. It is J-shaped when it is empty.
- * The stomach has the following **features:**
 - 1- It has **2 ends:** **Cardiac** end where the esophagus ends and **pyloric** end where the duodenum begins.
 - 2- It has **2 borders:** **Lesser curvature** (right border) and **greater curvature** (left border).
 - 3- It has **2 surfaces:** Anterior and posterior.
 - 4- It has **3 parts:**
 - a- **Fundus:** the part lying above the level of the cardia.
 - b- **Body.**
 - c- **Pyloric part:** consists of 3 parts: Antrum, canal and pyloric sphincter.





Small Intestine

* It consists of **3 parts**:

A- Duodenum

- The fixed part of the small intestine.
- It is C-shaped tube, 25 cm in length with its concave part directed to the left side and is occupied by the head of the pancreas.
- It is formed of 4 parts.

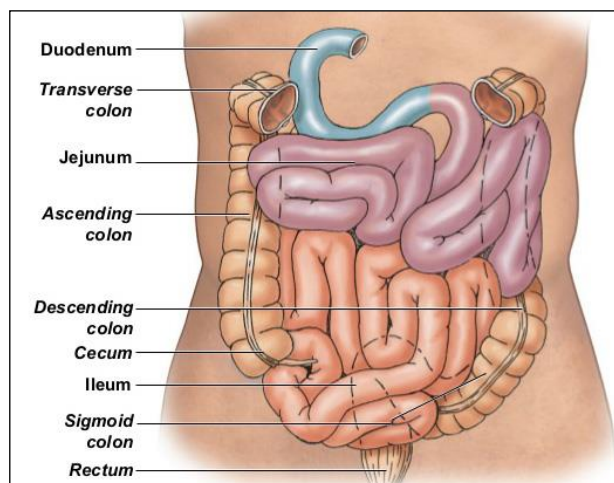
B- Jejunum.

C- Ileum.

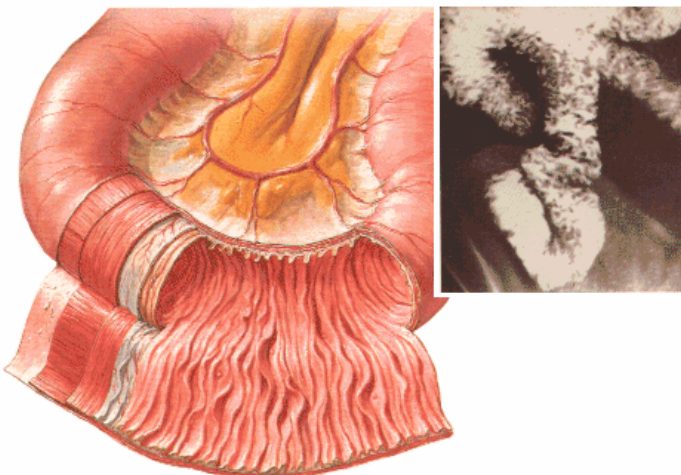
- Both jejunum and ileum are **mobile** part of small intestine.

- It is 6-meters in length.
- They are attached to the posterior wall of the abdomen by a fold of peritoneum called **mesentery** and are freely mobile.
- It **ends** by opening into the cecum (first part of large intestine)

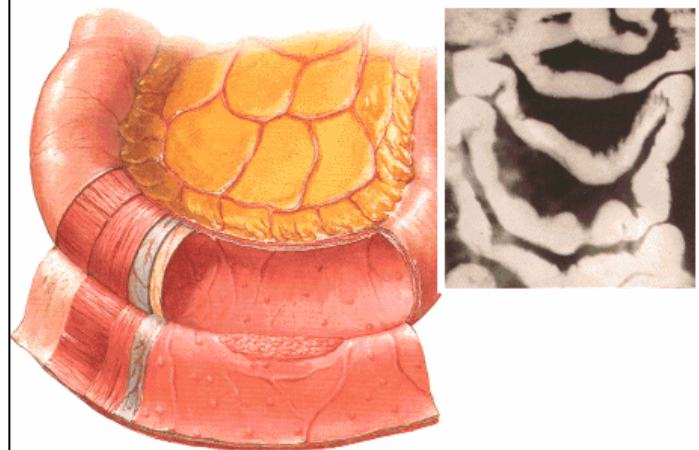
Jejunum	Ileum
• Lies above and to left in the abdominal cavity. • Forms the proximal 2/5 of the total length. • It has wide lumen and thick wall.	• Lies below and to right in the abdominal cavity. • Forms the distal 3/5 of the total length. • It has narrow lumen and thin wall.

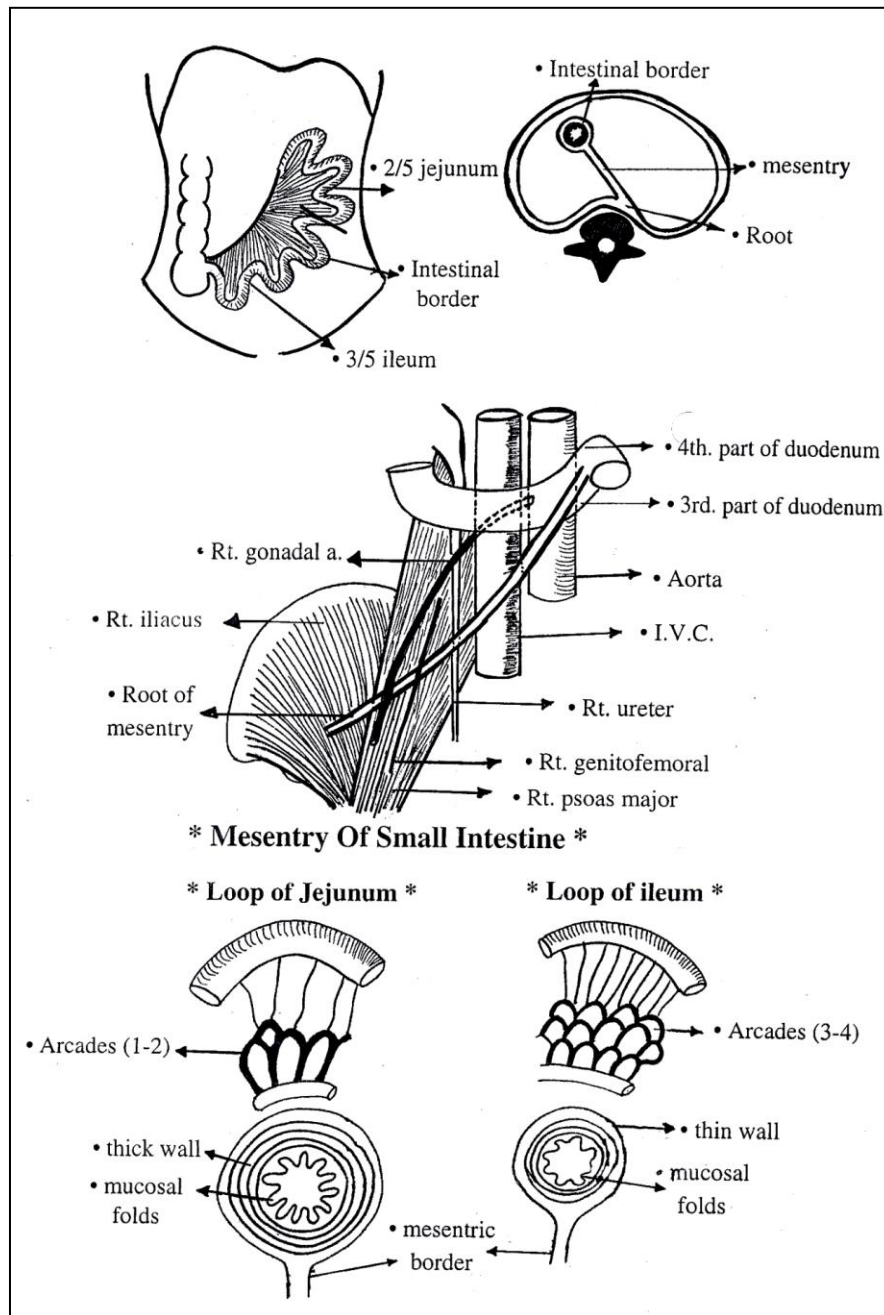


Mucosa and Musculature of Jejunum



Mucosa and Musculature of Ileum





Large Intestine

* It is 1.5 meter in length, wider lumen and has **3 main characters:**

1-Taenia coli: 3 bands of longitudinal smooth muscle fibers.

2- Appendices epiploicae: many small sacs containing fat.

3-Saculations: sacs like dilatations in the wall of the colon

* These characters are absent in the appendix and rectum.

* **Parts:**

1- Caecum: It lies in the lower right part of the abdominal cavity. It has lower blind end and it is continuous with ascending colon. The lower end of the ileum and the appendix open into it.

2- Vermiform appendix: It is variable in length and in position and it opens into the caecum below the opening of the ileum.

3- Ascending colon.

4- Right colic flexure.

5- Transverse colon.

6- Left colic flexure.

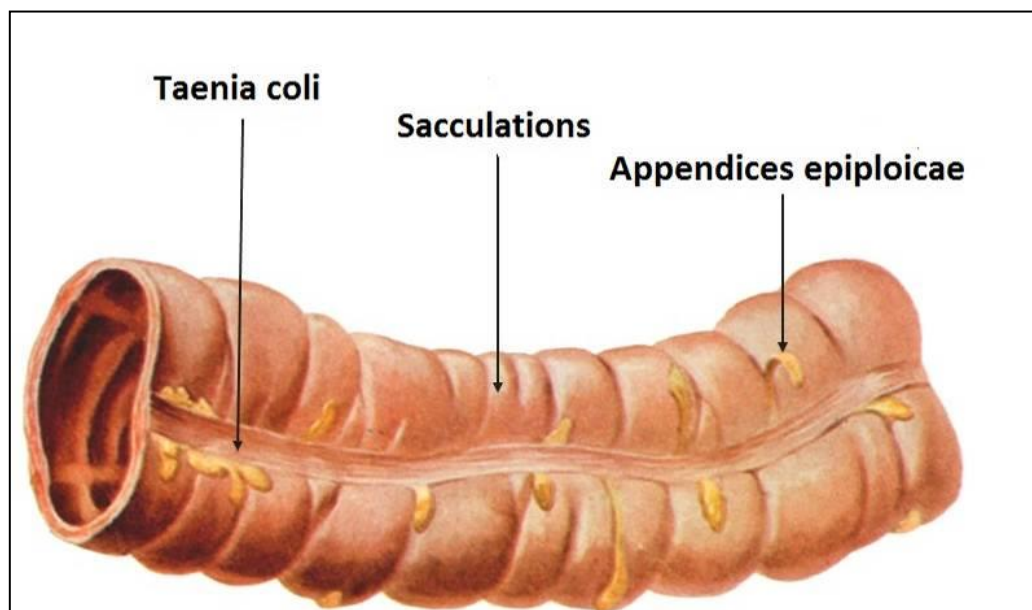
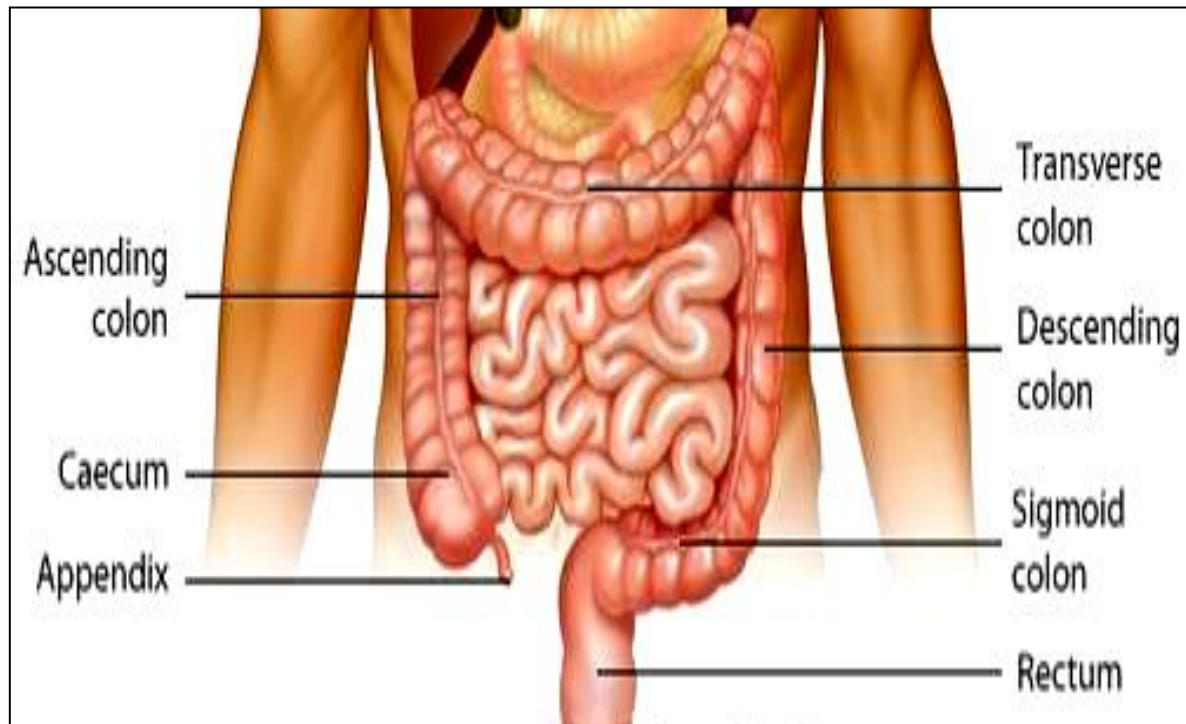
7- Descending colon.

8- Sigmoid (pelvic) colon.

9- Rectum.

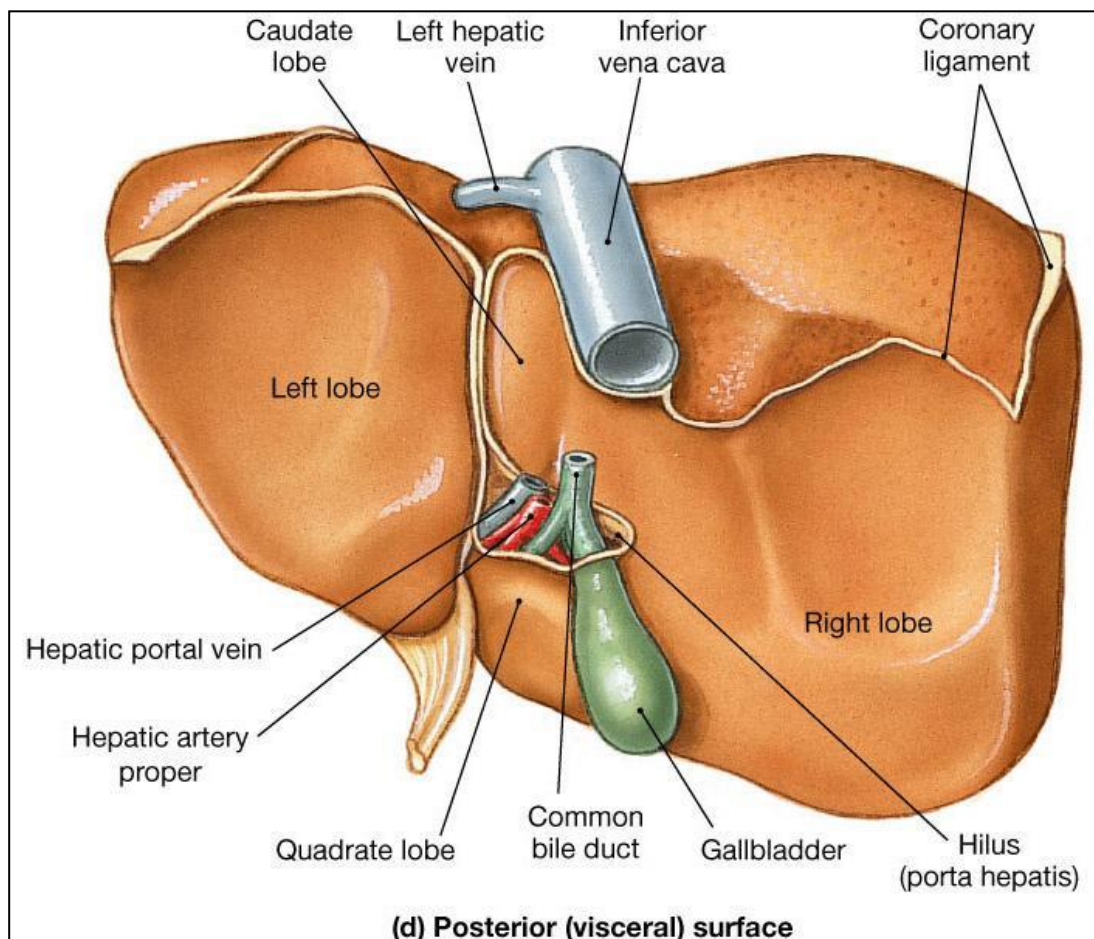
10- Anal canal which terminates below at the **external anal orifice**. It is directed downwards and backwards.

Large intestine



Liver

- The liver is the largest gland & largest single solid organ in the body
- It is **wedge-shaped** organ which has **5 surfaces**; right lateral surface (base), anterior, posterior, superior and inferior surfaces.
- It **lies** in the upper part of abdominal cavity below the diaphragm.
- **It consists of** large right lobe and small left lobe.
- Its hilum (**porta hepatis**) through which the common hepatic bile duct, the hepatic artery and portal vein pass.
- Its inferior surface has fossa for gall bladder.

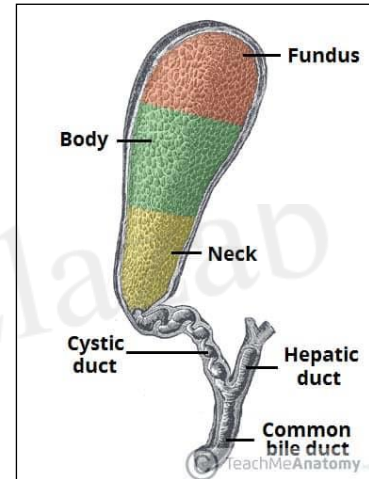


Biliary System

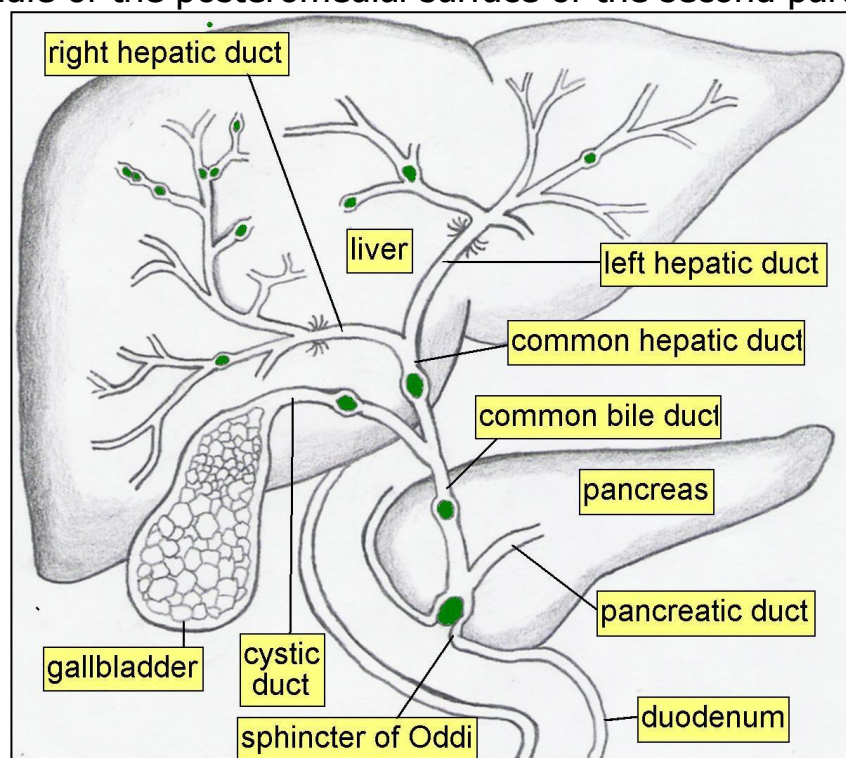
1- Liver cells secrete the bile and collected through the right and left hepatic ducts which unite together forming the common hepatic duct.

2- Gallbladder

- A pear-shaped organ that lies in special fossa in the inferior surface of the right lobe of the liver.
- It is formed of fundus, body and neck.
- It receives, stores and concentrates the bile which is secreted from liver.
- The bile leaves the gallbladder through the cystic duct which unites with the common hepatic duct to form the common bile duct.



3- The common bile duct unites with the pancreatic duct to form single duct which pierces the wall of **second part of duodenum** and becomes dilated inside the wall to form **ampula of Vater** which opens into the middle of the posteromedial surface of the second part of the duodenum.



Pancreas

- It is a yellow lobulated gland, lies transversely on the posterior abdominal wall.
- It **extends** from the duodenum on the right side to the spleen on the left side behind the stomach.
- It **consists of** 4 parts: head, neck, body, tail.
- **Function**
 1. **Endocrine function**: It secretes 2 hormones which regulate the level of the blood glucose, they are:
 - a- Insulin hormone: lowers the blood glucose level.
 - b- Glucagon hormone: Elevates the blood glucose level.
 2. **Exocrine function**: secretes pancreatic enzymes in the pancreatic juice which helps the digestion of food.

The pancreatic juice passes through the pancreatic duct which unites with the common bile duct and opens together into the second part of duodenum.

